

Worked examples for Lecture 17: The spherical Earth

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These examples accompany Lecture 17. They are intended as practice in the concrete calculations that underlie the lecture, and are less demanding than the questions on the problem sets. Throughout, b denotes the radius of the earth model, $\alpha(r)$ the P-wave speed, and $q = r \sin \theta / \alpha$ the ray parameter conserved along each ray. Numerical results were checked with the scripts in the python directory.

Example 1: The homogeneous sphere. Consider a spherically symmetric earth model of radius b in which the P-wave speed α is constant. For a ray leaving the surface with ray parameter q , evaluate the lecture's integrals for the epicentral angle Δ and the travel time T in closed form. Check the results against the elementary geometry of a chord, and verify directly that $dT/d\Delta = q$.

Solution. When α is constant the turning-point condition $q = r_t / \alpha(r_t)$ gives $r_t = \alpha q$, and the integral for the epicentral angle becomes

$$\Delta = \int_{\alpha q}^b \frac{2\alpha q}{r \sqrt{r^2 - \alpha^2 q^2}} dr. \quad (1)$$

Writing $a = \alpha q$ for the moment, a direct differentiation shows that

$$\frac{d}{dr} \arccos\left(\frac{a}{r}\right) = \frac{a}{r^2} \frac{1}{\sqrt{1 - a^2/r^2}} = \frac{a}{r \sqrt{r^2 - a^2}}, \quad (2)$$

so the integrand in eq. (1) is an exact derivative and

$$\Delta = 2 \left[\arccos\left(\frac{\alpha q}{r}\right) \right]_{\alpha q}^b = 2 \arccos\left(\frac{\alpha q}{b}\right), \quad (3)$$

the contribution from the lower limit vanishing because $\arccos 1 = 0$. In the same way, because $d\sqrt{r^2 - a^2}/dr = r/\sqrt{r^2 - a^2}$, the travel-time integral gives

$$T = \int_{\alpha q}^b \frac{2r}{\alpha \sqrt{r^2 - \alpha^2 q^2}} dr = \frac{2}{\alpha} \sqrt{b^2 - \alpha^2 q^2}. \quad (4)$$

We can check these results against elementary geometry. When α is constant the second of Hamilton's equations gives $d\mathbf{p}/d\sigma = \mathbf{0}$, so the rays are straight lines, and a ray between two surface points separated by the epicentral angle Δ is a chord of length $2b \sin(\Delta/2)$. Its travel time is therefore

$$T = \frac{2b}{\alpha} \sin \frac{\Delta}{2}. \quad (5)$$

From eq. (3) we have $\alpha q/b = \cos(\Delta/2)$, and hence $\sqrt{b^2 - \alpha^2 q^2} = b \sin(\Delta/2)$, which shows that eq. (4) agrees with eq. (5). The take-off angle is also consistent with the geometry: at the surface $q = b \sin \theta_0 / \alpha$, so $\sin \theta_0 = \cos(\Delta/2)$ and the chord makes an angle $\pi/2 - \Delta/2$ with the inward radius, exactly the base angle of the isosceles triangle formed by the chord and the two radii.

Finally, differentiating eq. (5) with respect to Δ and using $\cos(\Delta/2) = \alpha q/b$,

$$\frac{dT}{d\Delta} = \frac{b}{\alpha} \cos \frac{\Delta}{2} = q. \quad (6)$$

Note that Δ decreases as q increases, since from eq. (3) $d\Delta/dq = -2\alpha/\sqrt{b^2 - \alpha^2 q^2} < 0$: rays that leave the surface closer to the horizontal (larger q) travel less far. Consequently $d^2T/d\Delta^2 = dq/d\Delta < 0$, and the travel-time curve is concave, as are the observed curves shown in the lecture. The result $dT/d\Delta = q$ is not special to the homogeneous sphere, as the next example shows.

Example 2: The Benndorf relation. Let $\alpha(r)$ be such that $r/\alpha(r)$ increases monotonically with r , so that every ray leaving the surface with $0 < q < b/\alpha(b)$ has a unique turning point $r_t(q)$ determined by $q = r_t/\alpha(r_t)$. Starting from the lecture's integrals for $T(q)$ and $\Delta(q)$, show that

$$\frac{dT}{dq} = q \frac{d\Delta}{dq},$$

and hence that $dT/d\Delta = q$ along the travel-time curve. Explain how this allows the ray parameter, and hence the P-wave speed at the turning point, to be read off an observed travel-time curve.

Solution. Both T and Δ depend on q through the integrand and through the lower limit $r_t(q)$, and the integrands are singular at r_t . It is easier to work with the combination $\tau(q) = T - q\Delta$. Subtracting q times the Δ integral from the T integral and placing the two terms over a common denominator,

$$\tau(q) = T - q\Delta = \int_{r_t}^b \frac{2(r^2 - \alpha^2 q^2)}{\alpha r \sqrt{r^2 - \alpha^2 q^2}} dr = \int_{r_t}^b \frac{2}{\alpha r} \sqrt{r^2 - \alpha^2 q^2} dr. \quad (7)$$

This integrand is continuous and vanishes at the lower limit, because $r_t = \alpha(r_t)q$. Differentiating with respect to q using Leibniz's rule, the term arising from the motion of the lower limit is proportional to the integrand evaluated at r_t and so is zero, while differentiating under the integral sign gives

$$\frac{d\tau}{dq} = \int_{r_t}^b \frac{2}{\alpha r} \frac{-\alpha^2 q}{\sqrt{r^2 - \alpha^2 q^2}} dr = - \int_{r_t}^b \frac{2\alpha q}{r \sqrt{r^2 - \alpha^2 q^2}} dr = -\Delta(q). \quad (8)$$

The differentiated integrand is singular at r_t , but the singularity is of inverse square-root type and is integrable, and the interchange of differentiation and integration is legitimate. On the other hand, differentiating the definition $\tau = T - q\Delta$ directly,

$$\frac{d\tau}{dq} = \frac{dT}{dq} - \Delta - q \frac{d\Delta}{dq}, \quad (9)$$

and comparison with eq. (8) gives

$$\frac{dT}{dq} = q \frac{d\Delta}{dq}. \quad (10)$$

Wherever $d\Delta/dq \neq 0$ we can regard T as a function of Δ along the travel-time curve, and dividing eq. (10) by $d\Delta/dq$ yields

$$\frac{dT}{d\Delta} = q, \quad (11)$$

which is known as the Benndorf relation. Example 1 verified it for the homogeneous sphere.

The relation has a simple physical meaning. At the surface $q = b \sin \theta_0 / \alpha(b)$, so q/b is the horizontal component of the slowness vector of the emerging wavefront; equivalently the wavefront sweeps along the surface with apparent speed $b d\Delta/dT = b/q$. Its importance for the determination of Earth structure is the following. Suppose that we have measured the travel-time curve $T(\Delta)$ of a phase such as P. Its slope at a distance Δ gives the ray parameter $q(\Delta)$ of the ray emerging there, without any knowledge of the velocity model. The turning-point condition then tells us that

$$\frac{r_t}{\alpha(r_t)} = q(\Delta) \quad (12)$$

for the deepest point of that ray. A single number cannot fix both r_t and $\alpha(r_t)$, but the missing information is contained in the way q varies with Δ . Indeed, the integral in eq. (8) is of Abel type, and it can be inverted to give the turning radius r_1 of the ray with parameter q_1 arriving at Δ_1 as

$$\ln \frac{b}{r_1} = \frac{1}{\pi} \int_0^{\Delta_1} \cosh^{-1} \left[\frac{q(\Delta)}{q_1} \right] d\Delta, \quad (13)$$

after which $\alpha(r_1) = r_1/q_1$. This is the Herglotz–Wiechert formula, and it was the basis of the earliest determinations of the Earth's radial velocity structure. Its derivation requires Δ to be a monotonic function of q , which fails when r/α is not monotonic, for instance across a low-velocity zone, and this is one reason why the ad hoc methods mentioned in the lecture were needed in practice. For the homogeneous sphere the right-hand side of eq. (13) can be evaluated numerically using $q(\Delta) = (b/\alpha) \cos(\Delta/2)$, and it reproduces $\ln(b/r_1) = -\ln \cos(\Delta_1/2)$, as required by $r_1 = \alpha q_1$.

Example 3: The flat-earth transformation. Given a spherically symmetric model $\alpha(r)$, define a depth coordinate z and a transformed wave speed $\alpha_f(z)$ by

$$z = b \ln \frac{b}{r}, \quad \alpha_f(z) = \frac{b}{r} \alpha(r),$$

and set $q_f = q/b$. Show that the spherical integrals for T and Δ become the integrals for a horizontally stratified half-space obtained in Problem 7 of the first problem set, with q_f as the ray parameter and horizontal distance $X = b\Delta$. Use the transformation to find $\Delta(q)$ and $T(q)$ in closed form for the power-law model

$$\alpha(r) = \alpha_0 \left(\frac{b}{r} \right)^\nu, \quad \nu > -1,$$

and comment on the cases $\nu = 0$, $\nu = 1$ and $\nu \rightarrow -1$.

Solution. From $z = b \ln(b/r)$ we have $r = b e^{-z/b}$, so that $dr/r = -dz/b$, and the range $r_t \leq r \leq b$ maps to $0 \leq z \leq z_t$ with $z_t = b \ln(b/r_t)$, the orientation of the interval being reversed. The definitions of α_f and q_f have been arranged so that

$$\frac{\alpha(r)q}{r} = \frac{b}{r}\alpha(r)\frac{q}{b} = \alpha_f(z)q_f, \quad (14)$$

which is just the statement that $\sin \theta$ takes the same value at corresponding points of the two models. Dividing the numerator and denominator of the lecture's Δ integrand by r and using eq. (14),

$$\Delta = \int_{r_t}^b \frac{2\alpha q/r}{\sqrt{1 - \alpha^2 q^2/r^2}} \frac{dr}{r} = \frac{1}{b} \int_0^{z_t} \frac{2q_f \alpha_f}{\sqrt{1 - q_f^2 \alpha_f^2}} dz = \frac{X}{b}, \quad (15)$$

where X is precisely the horizontal distance integral of Problem 7. For the travel time we use $1/\alpha = (b/r)/\alpha_f$ together with $(b/r) dr = -dz$ to obtain

$$T = \int_{r_t}^b \frac{2}{\alpha \sqrt{1 - \alpha^2 q^2/r^2}} dr = \int_0^{z_t} \frac{2}{\alpha_f \sqrt{1 - q_f^2 \alpha_f^2}} dz, \quad (16)$$

which is the flat-earth travel time. The turning-point conditions also correspond, because $q = r_t/\alpha(r_t)$ is equivalent to $q_f \alpha_f(z_t) = 1$. The transformation is exact for the kinematics of rays in any spherically symmetric model, and it allows methods developed for horizontally stratified media to be applied to the Earth.

For the power-law model the transformed wave speed is

$$\alpha_f(z) = \frac{b}{r} \alpha_0 \left(\frac{b}{r} \right)^\nu = \alpha_0 e^{(1+\nu)z/b} = \alpha_0 e^{z/h}, \quad h = \frac{b}{1+\nu}, \quad (17)$$

so that every power law becomes an exponential flat-earth model, with scale height $h > 0$ precisely when $\nu > -1$. In the flat-earth integrals we substitute $u = q_f \alpha_f(z) = q_f \alpha_0 e^{z/h}$, for which $dz = h du/u$, and u runs from $u_0 = q_f \alpha_0 = \alpha_0 q/b$ at the surface to 1 at the turning depth. Then

$$X = 2h \int_{u_0}^1 \frac{du}{\sqrt{1 - u^2}} = 2h \arccos u_0, \quad (18)$$

and, using $1/\alpha_f = q_f/u$ together with $d(-\sqrt{1 - u^2}/u)/du = 1/(u^2 \sqrt{1 - u^2})$,

$$T = 2h q_f \int_{u_0}^1 \frac{du}{u^2 \sqrt{1 - u^2}} = 2h q_f \left[-\frac{\sqrt{1 - u^2}}{u} \right]_{u_0}^1 = \frac{2h}{\alpha_0} \sqrt{1 - u_0^2}. \quad (19)$$

Transforming back with $\Delta = X/b$, $h = b/(1 + \nu)$ and $u_0 = \alpha_0 q/b$, we arrive at

$$\Delta(q) = \frac{2}{1 + \nu} \arccos \left(\frac{\alpha_0 q}{b} \right), \quad T(q) = \frac{2}{(1 + \nu) \alpha_0} \sqrt{b^2 - \alpha_0^2 q^2}, \quad (20)$$

with turning radius $r_t = b e^{-z_t/b} = b (\alpha_0 q / b)^{1/(1+\nu)}$, which agrees with the direct solution of $q = r_t / \alpha(r_t)$. Eliminating q , the travel-time curve is

$$T(\Delta) = \frac{2b}{(1+\nu)\alpha_0} \sin \frac{(1+\nu)\Delta}{2}, \quad (21)$$

and differentiating gives $dT/d\Delta = (b/\alpha_0) \cos[(1+\nu)\Delta/2] = q$, in agreement with the Benndorf relation. These closed forms were checked against numerical quadrature of the original spherical integrals; for example with $b = 6371$ km, $\alpha_0 = 8$ km s⁻¹, $\nu = 1$ and $q = 300$ s rad⁻¹ we find $\Delta = 67.87^\circ$, $T = 737.7$ s and $r_t = 3910$ km.

For $\nu = 0$ we recover the results of Example 1. For $\nu = 1$, in which $\alpha \propto 1/r$, a ray with a given take-off angle subtends exactly half the epicentral angle, and takes exactly half the time, of the corresponding ray in the homogeneous sphere with $\alpha = \alpha_0$: the increase of wave speed with depth bends the ray back to the surface more quickly. Since $\alpha \rightarrow \infty$ as $r \rightarrow 0$, this model is only sensible for a shell such as the mantle, and eq. (20) should be used for rays whose turning radius lies within that shell; with the numbers above the surface speed of 8 km s⁻¹ increases to 14.6 km s⁻¹ at $r = 3480$ km. As $\nu \rightarrow -1$ the scale height $h \rightarrow \infty$, and for $\nu = -1$ the flat-earth model is homogeneous, so that the transformed rays are straight lines that never turn. In the sphere $\alpha \propto r$, the quantity $\sin \theta = \alpha q / r$ is constant along each ray, and the rays are logarithmic spirals that wind into the centre. More generally, a ray turns only where r/α increases outwards, that is where $d\alpha/dr < \alpha/r$; a wave speed that increases with depth faster than this critical rate, as happens for $\nu < -1$, produces rays that never return to the surface.

Example 4: Hypocentre location by linearised least squares. Consider a homogeneous half-space with known P-wave speed α , and use coordinates in which x_3 measures depth below the free surface. Seismometers at surface positions $\mathbf{x}_i$ ($i = 1, \dots, N_s$) record the first arrival times t_i from an event with unknown hypocentre $\mathbf{x}_h$ and origin time t_h . (a) Show that

$$\frac{\partial T(\mathbf{x}_i, \mathbf{x}_h)}{\partial \mathbf{x}_h} = -\frac{\mathbf{x}_i - \mathbf{x}_h}{\alpha \|\mathbf{x}_i - \mathbf{x}_h\|},$$

and interpret the result. (b) Set up the Gauss–Newton iteration for minimising the least-squares misfit $J(\mathbf{x}_h, t_h)$ of the lecture. (c) Show that if all the seismometers lie at the same horizontal distance from the epicentre, the normal matrix is singular, and identify the combination of parameters that cannot be determined. (d) Illustrate the iteration numerically with $\alpha = 6$ km s⁻¹ and five stations.

Solution. (a) In a homogeneous medium rays are straight lines, so $T(\mathbf{x}_i, \mathbf{x}_h) = \|\mathbf{x}_i - \mathbf{x}_h\|/\alpha$. Using $\partial \|\mathbf{y}\|/\partial \mathbf{y} = \mathbf{y}/\|\mathbf{y}\|$ with $\mathbf{y} = \mathbf{x}_i - \mathbf{x}_h$, and the chain rule with $\partial \mathbf{y}/\partial \mathbf{x}_h = -\mathbf{I}$,

$$\frac{\partial T(\mathbf{x}_i, \mathbf{x}_h)}{\partial \mathbf{x}_h} = -\frac{1}{\alpha} \hat{\mathbf{n}}_i, \quad \hat{\mathbf{n}}_i = \frac{\mathbf{x}_i - \mathbf{x}_h}{\|\mathbf{x}_i - \mathbf{x}_h\|}. \quad (22)$$

Here $\hat{\mathbf{n}}_i/\alpha$ is the slowness vector $\mathbf{p}$ of the ray at the source, so the derivative is minus the slowness vector at the source. This is a general property of ray theory: the travel time increases at the rate $\|\mathbf{p}\| = 1/\alpha$ when the source is moved away from the receiver

along the ray, and does not change to first order when it is moved perpendicular to the ray. Moving the source has the same effect as moving the receiver in the opposite direction, which is why the sign is negative. The derivative with respect to the origin time is simply $\partial(t_h + T)/\partial t_h = 1$.

(b) Collect the unknowns into $\mathbf{m} = (\mathbf{x}_h, t_h)$ and define the weighted residuals $r_i(\mathbf{m}) = [t_h + T(\mathbf{x}_i, \mathbf{x}_h) - t_i]/\sigma_i$, so that $J = \sum_i r_i^2$. Given a current estimate $\mathbf{m}$, the residuals for a nearby model are linearised as

$$r_i(\mathbf{m} + \delta\mathbf{m}) \approx r_i(\mathbf{m}) + \sum_{j=1}^4 G_{ij} \delta m_j, \quad G_{ij} = \frac{1}{\sigma_i} \frac{\partial(t_h + T_i)}{\partial m_j}, \quad (23)$$

where by eq. (22) the i th row of the $N_s \times 4$ matrix $\mathbf{G}$ is $\sigma_i^{-1}(-\hat{\mathbf{n}}_i^T/\alpha, 1)$. Minimising the resulting quadratic approximation to J with respect to $\delta\mathbf{m}$ gives the normal equations

$$\mathbf{G}^T \mathbf{G} \delta\mathbf{m} = -\mathbf{G}^T \mathbf{r}, \quad (24)$$

after which $\mathbf{m}$ is replaced by $\mathbf{m} + \delta\mathbf{m}$ and the process is repeated until $\delta\mathbf{m}$ is negligible. Each iteration requires the travel times and their derivatives for the current hypocentre, which in a realistic earth model are obtained by ray tracing. At least four stations in general position are needed for the 4×4 normal matrix $\mathbf{G}^T \mathbf{G}$ to be invertible.

(c) Let the hypocentre lie at depth h beneath the origin of horizontal coordinates, and let the i th station lie at horizontal distance d_i , so that $D_i = \|\mathbf{x}_i - \mathbf{x}_h\| = \sqrt{d_i^2 + h^2}$. Because the stations are on the surface, the depth component of $\hat{\mathbf{n}}_i$ is $-h/D_i$, and the third and fourth entries of the i th row of $\mathbf{G}$ are

$$G_{i3} = \frac{1}{\sigma_i} \frac{h}{\alpha D_i}, \quad G_{i4} = \frac{1}{\sigma_i}. \quad (25)$$

If all the stations are at the same horizontal distance d , then $D_i = D$ for every i and the third column of $\mathbf{G}$ is $h/(\alpha D)$ times the fourth. The vector $\mathbf{v} = (0, 0, 1, -h/(\alpha D))^T$ then satisfies $\mathbf{G}\mathbf{v} = \mathbf{0}$, so the normal matrix has a null vector and eq. (24) cannot be solved uniquely. Physically, deepening the source by δh lengthens every ray by the same amount and delays every arrival by $h \delta h/(\alpha D)$, which can be undone exactly by making the origin time earlier by the same amount. In this configuration the trade-off is in fact exact rather than merely linear, since all the predicted arrival times depend on h and t_h only through the single combination $t_h + \sqrt{d^2 + h^2}/\alpha$. The depth of an earthquake is resolved only by stations at a range of epicentral distances, for which the take-off angles differ, and in practice depth is the most poorly determined of the four hypocentral parameters.

(d) We take stations at the horizontal positions (30, 0), (0, 40), (−25, 15), (−10, −35) and (20, −20) km, a true hypocentre at (3, −2, 12) km with $t_h = 0$, and equal uncertainties $\sigma_i = 0.1$ s. The synthetic arrival times are 4.936, 7.297, 5.814, 6.241 and 4.586 s, with no noise added so that the exact answer is known. Starting from $\mathbf{x}_h = (0, 0, 5)$ km and $t_h = 1$ s, the Gauss–Newton iteration gives

iteration	$x_{h,1}$ (km)	$x_{h,2}$ (km)	$x_{h,3}$ (km)	t_h (s)
0	0.000	0.000	5.000	1.0000
1	2.795	−1.868	13.377	0.0966
2	3.004	−2.001	12.034	0.0017
3	3.000	−2.000	12.000	0.0000

with the misfit falling from $J = 370$ to 18 , 6.7×10^{-3} and 3.4×10^{-9} . The convergence is quadratic once the iterate is close to the solution, as expected for a Gauss–Newton method applied to data that can be fitted exactly. If instead the same five stations are placed at azimuths $0^\circ, 72^\circ, \dots, 288^\circ$ on a circle of radius 30 km about the epicentre of a source at 12 km depth, the eigenvalues of $\mathbf{G}^T \mathbf{G}$ are found to be 0, 5.99, 5.99 and 502 (in units of s^{-2} with distances in kilometres), the null vector being proportional to $(0, 0, 1, -0.0619)$ in agreement with $h/(\alpha D) = 12/(6 \times 32.31) = 0.0619$ from eq. (25).

Example 5: The P-wave shadow zone. Consider a two-layer earth model of radius $b = 6371$ km consisting of a homogeneous mantle with $\alpha_m = 13 \text{ km s}^{-1}$ above a homogeneous core of radius $c = 3480$ km with $\alpha_c = 8 \text{ km s}^{-1}$. For a source at the surface: (a) find the epicentral angle Δ_1 of the mantle ray that grazes the core; (b) derive $\Delta(q)$ and $T(q)$ for rays refracted through the core, using the fact that the ray parameter is continuous across the core–mantle boundary (CMB); (c) find the smallest angular distance at which core-refracted P-waves arrive, and hence the extent of the P-wave shadow zone. Compare with the observed shadow zone, which extends from about 103° to 142° .

Solution. (a) In the homogeneous mantle rays are straight lines, and by Example 1 a ray with parameter q would turn at radius $\alpha_m q$. It remains in the mantle if $\alpha_m q > c$, and it grazes the core when $q = q_1 = c/\alpha_m = 267.7 \text{ s rad}^{-1}$. Its epicentral angle follows from eq. (3),

$$\Delta_1 = 2 \arccos \left(\frac{\alpha_m q_1}{b} \right) = 2 \arccos \frac{c}{b} = 113.8^\circ, \quad (26)$$

and its travel time from eq. (4) is $T_1 = 2\sqrt{b^2 - c^2}/\alpha_m = 821 \text{ s}$. Mantle P-waves therefore arrive at all distances $0 < \Delta < 113.8^\circ$, and at no greater distance, as the grazing ray is the one that leaves the surface most steeply, with $\sin \theta_0 = c/b$, that is at 33.1° from the vertical, without entering the core.

(b) A ray with $q < q_1$ reaches the CMB. The down-going mantle leg from $r = b$ to $r = c$ subtends an angle obtained from the antiderivative in eq. (2),

$$\Delta_m(q) = \int_c^b \frac{\alpha_m q}{r \sqrt{r^2 - \alpha_m^2 q^2}} dr = \arccos \left(\frac{\alpha_m q}{b} \right) - \arccos \left(\frac{\alpha_m q}{c} \right). \quad (27)$$

At the CMB the ray is refracted. Since $q = r \sin \theta / \alpha$ and $r = c$ on both sides of the boundary, continuity of q is the statement that $\sin \theta_c / \alpha_c = \sin \theta_m / \alpha_m$, which is Snell's law; it expresses the fact that the wavefront, and hence the phase along the boundary, is continuous across it. Because $\alpha_c < \alpha_m$ the ray is bent towards the radial direction on entering the core, and there is no critical angle: every mantle ray that reaches the CMB is transmitted. Inside the homogeneous core the ray is again a straight line with the same q , and its turning radius $\alpha_c q$ is automatically less than c . By eq. (3) with b replaced by c , the core leg subtends $2 \arccos(\alpha_c q / c)$, and adding the symmetric up-going mantle leg,

$$\Delta(q) = 2 \arccos \left(\frac{\alpha_m q}{b} \right) - 2 \arccos \left(\frac{\alpha_m q}{c} \right) + 2 \arccos \left(\frac{\alpha_c q}{c} \right), \quad 0 \leq q < q_1. \quad (28)$$

The travel time is assembled in the same way from eq. (4),

$$T(q) = \frac{2}{\alpha_m} \left[\sqrt{b^2 - \alpha_m^2 q^2} - \sqrt{c^2 - \alpha_m^2 q^2} \right] + \frac{2}{\alpha_c} \sqrt{c^2 - \alpha_c^2 q^2}, \quad (29)$$

and one may check, either directly or numerically, that $dT/dq = q d\Delta/dq$ continues to hold, so that the Benndorf relation survives the presence of the discontinuity.

(c) At the two ends of the range of q we have

$$\lim_{q \rightarrow q_1} \Delta(q) = \Delta_1 + 2 \arccos \frac{\alpha_c}{\alpha_m} = 113.8^\circ + 104.0^\circ = 217.8^\circ, \quad \Delta(0) = 180^\circ, \quad (30)$$

the first because a ray entering the core at grazing incidence is refracted to an angle θ_c from the radial direction given by $\sin \theta_c = \alpha_c/\alpha_m$, that is 38.0° , and then crosses the core along a long chord, the second because the vertical ray passes through the centre. As q decreases from q_1 the second term in eq. (28) grows rapidly, with $d\Delta/dq \rightarrow +\infty$ at q_1 , and Δ falls steeply before rising again towards 180° . Minimising eq. (28) numerically, the stationary point is at $q_B = 157.6 \text{ s rad}^{-1}$ where $\Delta_B = 172.1^\circ$. This is a caustic, at which neighbouring rays cross and ray theory predicts an infinite amplitude, and it is the analogue for this model of the caustic responsible for the strong PKP arrivals observed near 144° in the Earth.

The stationary point is not, however, the nearest point to the source at which core-refracted energy arrives. An epicentral angle $\Delta > 180^\circ$ means that the ray has passed beyond the antipode, and the station lies at the angular distance $360^\circ - \Delta$ from the source on the far side. The relevant quantity is therefore $\delta(q) = \min[\Delta(q), 360^\circ - \Delta(q)]$, and a numerical search over $0 \leq q < q_1$ shows that its infimum is attained in the grazing limit,

$$\delta_{\min} = 360^\circ - \lim_{q \rightarrow q_1} \Delta(q) = 360^\circ - 113.8^\circ - 104.0^\circ = 142.2^\circ. \quad (31)$$

Core-refracted P-waves thus arrive at angular distances between 142.2° and 180° ; between 172.1° and 180° three distinct core rays arrive at each distance, one from each monotonic branch of $\Delta(q)$, while between 142.2° and 172.1° there is a single arrival, from the branch with q close to q_1 . Combining this with part (a), no geometrical P-wave arrives in the interval

$$113.8^\circ < \delta < 142.2^\circ, \quad (32)$$

which is the P-wave shadow zone of this model. Figure 1 shows a fan of rays, the gap in surface coverage being clearly visible.

The far edge of the shadow zone agrees remarkably well with the observed value of 142° , although this is partly fortuitous: in the real Earth it is set by the PKP caustic rather than by the grazing ray, and its position depends on the details of the core velocity structure. The near edge is predicted at 113.8° rather than the observed 103° because the wave speed of the real mantle increases with depth. The mantle legs of the grazing ray are then curved back towards the surface and each subtends less than $\arccos(c/b) = 56.9^\circ$, and the whole of Oldham's shadow zone moves closer to the source. Within the shadow, seismograms are not silent: the core reflection PcP, waves diffracted around the CMB, and S-waves and surface waves are all recorded there, but the sharp, direct P-wave that dominates the first arrival at shorter distances is absent, and it was this absence that led Oldham to infer the existence of the core.

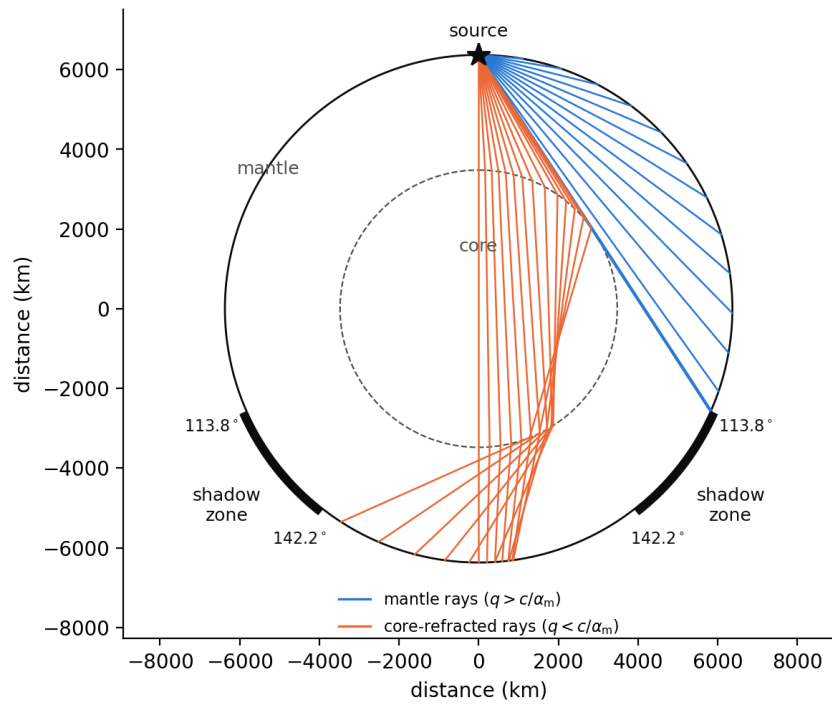


Fig. 1: Ray paths for a surface source in the two-layer model of Example 5. Mantle rays (blue) are straight lines that emerge at distances up to 113.8° ; rays that enter the core (orange) are refracted towards the radial direction at the CMB and emerge no closer to the source than 142.2° . The thick arcs mark the shadow zone on either side of the source.